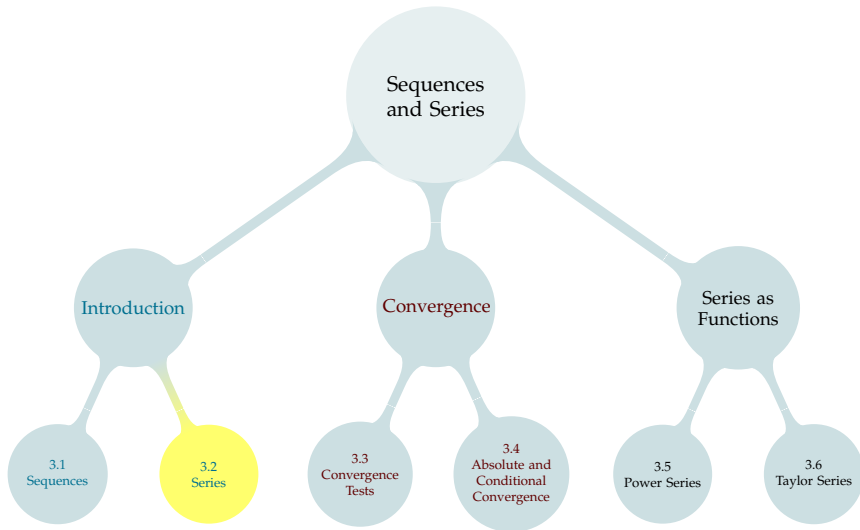


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SEQUENCES AND SERIES

A **sequence** is a list of numbers
A **series** is the sum of such a list.

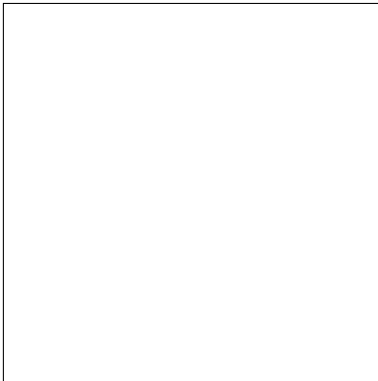
SEQUENCES AND SERIES

Sequence

List of numbers,
approaching

Series

Sum of numbers,
approaching



Square of Area 1

Recall:

$$\sum_{n=1}^5 \frac{1}{n^2} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2}$$

We informally interpret:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \frac{1}{6^2} + \frac{1}{7^2} + \frac{1}{8^2} + \frac{1}{9^2} + \frac{1}{10^2} + \cdots$$

(a more rigorous definition will be discussed soon)

Let a_n and b_n be sequences, and let C be a constant.

$$\sum_{n=1}^{\infty} (C \cdot a_n) =$$

$$\text{A. } \sum_{n=1}^{\infty} C \cdot \sum_{n=1}^{\infty} a_n$$

$$\text{B. } \sum_{n=1}^{\infty} C + \sum_{n=1}^{\infty} a_n$$

$$\text{C. } C \sum_{n=1}^{\infty} a_n$$

$$\text{D. } a_n \sum_{n=1}^{\infty} C$$

E. none of the above

Let a_n and b_n be sequences, and let C be a constant.

$$\sum_{n=1}^{\infty} (a_n + b_n) =$$

$$\text{A. } \sum_{n=1}^{\infty} a_n \cdot \sum_{n=1}^{\infty} b_n$$

$$\text{B. } \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n$$

C. $a_n + \sum_{n=1}^{\infty} b_n$

$$\text{D. } a_n \sum_{n=1}^{\infty} b_n$$

E. none of the above

Let a_n and b_n be sequences, and let C be a constant.

$$\sum_{n=1}^{\infty} (a_n)^C =$$

$$\text{A. } \sum_{n=1}^{\infty} C \cdot \sum_{n=1}^{\infty} a_n$$

B. $\left(\sum_{n=1}^{\infty} a_n\right)^C$

$$\text{C. } C^n \sum_{n=1}^{\infty} a_n$$

$$\text{D. } \sum_{n=1}^{\infty} C(a_n)^{C-1}$$

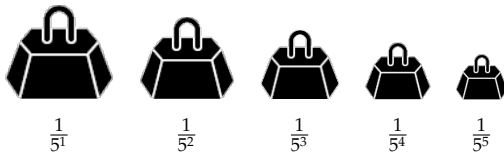
E. none of the above

What does it really mean to add up infinitely many things?

We need an unambiguous definition.

HOW CAN WE ADD UP INFINITELY MANY THINGS?

SEQUENCE OF PARTIAL SUMS



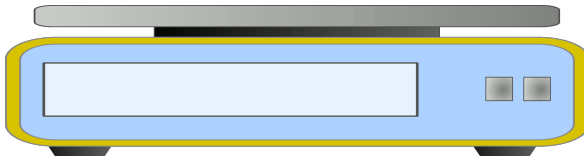
$$S_1 = 0.2000$$

$$S_2 = 0.2400$$

$$S_3 = 0.2480$$

$$S_4 = 0.2496$$

$$S_5 = 0.2499$$



Partial sums let us think about series (sums) using the tools we've developed for sequences (lists).

$$a_1 = \frac{1}{5} = 0.2$$

$$S_1 = 0.2$$

$$a_2 = \frac{1}{5^2} = 0.04$$

$$S_2 = 0.24$$

$$a_3 = \frac{1}{5^3} = 0.008$$

$$S_3 = 0.248$$

$$a_4 = \frac{1}{5^4} = 0.0016$$

$$S_4 = 0.2496$$

$$a_5 = \frac{1}{5^5} = 0.00032$$

$$S_5 = 0.24992$$

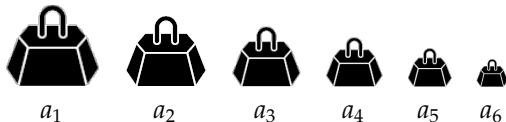
We define $\sum_{n=1}^{\infty} a_n = \lim_{N \rightarrow \infty} \sum_{n=1}^N a_n = \lim_{N \rightarrow \infty} S_N.$

$a_1 = \frac{1}{5} = 0.2$	$S_1 = 0.2$	$a_5 = \frac{1}{5^5} = 0.00032$	$S_5 = 0.24992$
$a_2 = \frac{1}{5^2} = 0.04$	$S_2 = 0.24$	$a_6 = \frac{1}{5^6} = 0.000064$	$S_6 = 0.249984$
$a_3 = \frac{1}{5^3} = 0.008$	$S_3 = 0.248$	$a_7 = \frac{1}{5^7} = 0.0000128$	$S_7 = 0.2499968$
$a_4 = \frac{1}{5^4} = 0.0016$	$S_4 = 0.2496$	$a_8 = \frac{1}{5^8} = 0.00000256$	$S_8 = 0.24999936$

From the sequence of partial sums, we guess

$$\sum_{n=1}^{\infty} = \lim_{N \rightarrow \infty} S_N =$$

NOTATION: $S_N = \sum_{n=1}^N a_n$



$$S_1 = a_1$$

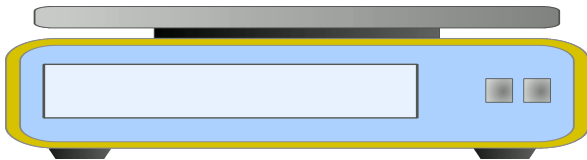
$$S_2 = a_1 + a_2$$

$$S_3 = a_1 + a_2 + a_3$$

$$S_4 = a_1 + \cdots + a_4$$

$$S_5 = a_1 + \cdots + a_5$$

$$S_6 = a_1 + \cdots + a_6$$



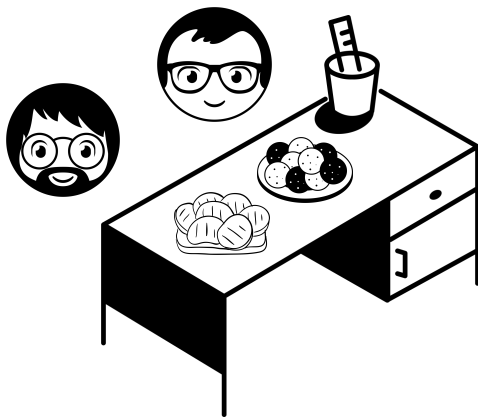
NOTATION PRACTICE

Suppose $\sum_{n=1}^{\infty} a_n$ has partial sums $S_N = \sum_{n=1}^N a_n = \frac{N}{N+1}$.

► Evaluate $\sum_{n=1}^{100} a_n$.

► Evaluate $\sum_{n=1}^{\infty} a_n$.

NOTATION PRACTICE

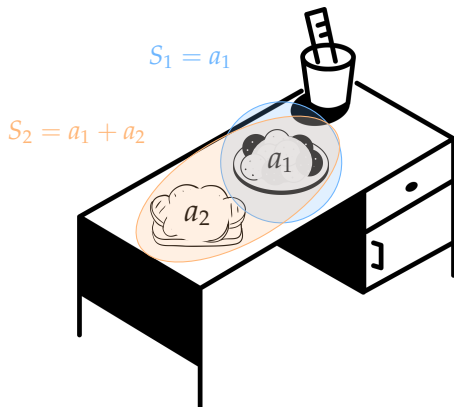


Andrew brings a plate of cookies to the professor's desk. When he puts them down, there are 10 cookies on the desk.

Then, Joel brings a plate of cookies. When he puts them down, there are 19 cookies on the desk.

How many cookies did each person bring?

NOTATION PRACTICE



Andrew brings a plate of cookies to the professor's desk. When he puts them down, there are 10 cookies on the desk.

Then, Joel brings a plate of cookies. When he puts them down, there are 19 cookies on the desk.

How many cookies did each person bring?

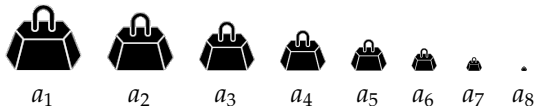
NOTATION PRACTICE

Suppose $\sum_{n=1}^{\infty} a_n$ has partial sums $S_N = \sum_{n=1}^N a_n = \frac{N}{N+1}$.

- Find a_1 .
- Give an explicit expression for a_n , when $n > 1$.

○○○○○○○○○○○○●○○○○○○○○○○○○○○

$$S_N = \sum_{n=1}^N a_n = \frac{N}{N+1}$$



$$S_1 = 1/(1+1)$$

$$S_2 = 2/(2+1)$$

$$S_3 = 3/(3+1)$$

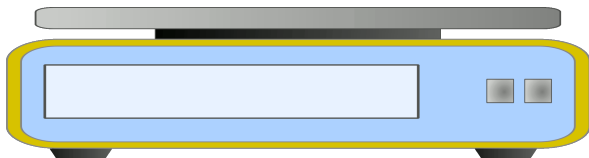
$$S_4 = 4/(4+1)$$

$$S_5 = 5/(5+1)$$

$$S_6 = 6/(6+1)$$

$$S_7 = 7/(7+1)$$

$$S_8 = 8/(8+1)$$



Definition

The N^{th} partial sum of the series $\sum_{n=1}^{\infty} a_n$ is the sum of its first N terms

$$S_N = \sum_{n=1}^N a_n.$$

The partial sums form a sequence $\{S_N\}_{N=1}^{\infty}$. If this sequence of partial sums converges $S_N \rightarrow S$ as $N \rightarrow \infty$ then we say that the series $\sum_{n=1}^{\infty} a_n$ converges to S and we write

$$\sum_{n=1}^{\infty} a_n = S$$

If the sequence of partial sums diverges, we say that the series diverges.

Geometric Series

Let a and r be two fixed real numbers with $a \neq 0$. The series

$$a + ar + ar^2 + ar^3 + \dots$$

is called the **geometric series** with first term a and ratio r .

We call r the *ratio* because it is the quotient of consecutive terms:

$$\frac{ar^{n+1}}{ar^n} = r$$

Another useful way of identifying geometric series is to determine whether all pairs of consecutive terms have the same ratio.

- ▶ Geometric: $1 + \frac{1}{5} + \frac{1}{5^2} + \frac{1}{5^3} + \frac{1}{5^4} + \dots$
- ▶ Geometric: $\sum_{n=0}^{\infty} \frac{1}{2^n}$
- ▶ Not geometric: $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \dots$

Consider the partial sum S_N of a geometric series:

$$S_N = a + ar + ar^2 + ar^3 + \cdots + ar^N$$

$$rS_N =$$

$$rS_N - S_N =$$

$$S_N(r - 1) = ar^{N+1} - a$$

If $r \neq 1$, then

$$S_N = \frac{ar^{N+1} - a}{r - 1} = a \frac{r^{N+1} - 1}{r - 1}$$

Geometric Series and Partial Sums

Let a and r be constants with $a \neq 0$, and let N be a natural number.

► If $r \neq 1$, then $a + ar + ar^2 + ar^3 + \cdots + ar^N = a \frac{r^{N+1} - 1}{r - 1}.$

► If $r = 1$, then $a + ar + ar^2 + ar^3 + \cdots + ar^N =$

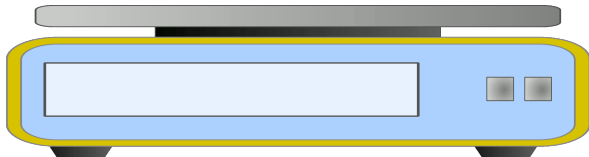
► If $|r| < 1$, then $\sum_{n=0}^{\infty} ar^n =$

► If $r = 1$, then $\sum_{n=0}^{\infty} ar^n$

► If $r = -1$, then $\sum_{n=0}^{\infty} ar^n$

► If $|r| > 1$, then $\sum_{n=0}^{\infty} ar^n$

$$\sum_{n=0}^{\infty} ar^n, r = 1, a \neq 0$$



$$S_0 = a$$

$$S_1 = 2a$$

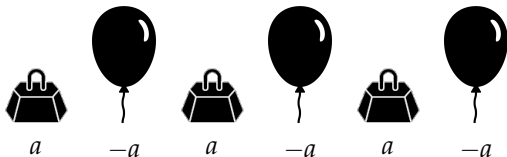
$$S_2 = 3a$$

$$S_3 = 4a$$

$$S_4 = 5a$$

$$S_5 = 6a$$

$$\sum_{n=0}^{\infty} ar^n, r = -1, a \neq 0$$



$$S_0 = a$$

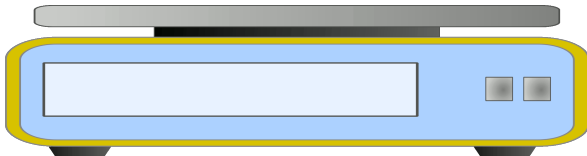
$$S_1 = 0$$

$$S_2 = a$$

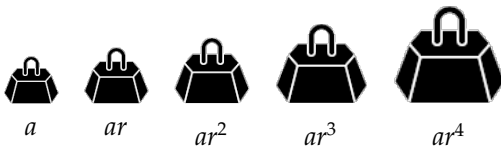
$$S_3 = 0$$

$$S_4 = a$$

$$S_5 = 0$$



$$\sum_{n=0}^{\infty} ar^n, r > 1, a \neq 0$$



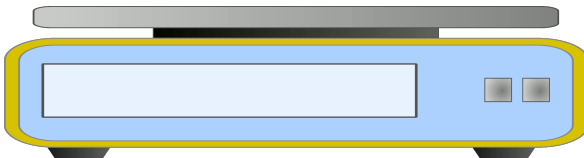
$$S_0 = a$$

$$S_1 = a \frac{r^2 - 1}{r - 1}$$

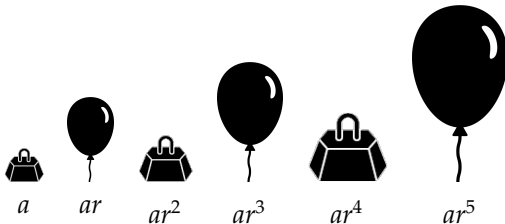
$$S_2 = a \frac{r^3 - 1}{r - 1}$$

$$S_3 = a \frac{r^4 - 1}{r - 1}$$

$$S_4 = a \frac{r^5 - 1}{r - 1}$$



$$\sum_{n=0}^{\infty} ar^n, r < -1, a \neq 0$$



$$S_0 = a$$

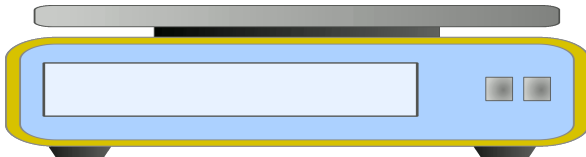
$$S_1 = a \frac{r^2 - 1}{r - 1}$$

$$S_2 = a \frac{r^3 - 1}{r - 1}$$

$$S_3 = a \frac{r^4 - 1}{r - 1}$$

$$S_4 = a \frac{r^5 - 1}{r - 1}$$

$$S_5 = a \frac{r^6 - 1}{r - 1}$$



GEOMETRIC SERIES

New bitcoins are produced when a particular type of computational problem is solved. Every time 210,000 solutions are found, the number of bitcoins each solution can produce is cut in half.

- ▶ Each of the first 210,000 solutions can produce 50 bitcoins.
- ▶ Each of the next 210,000 solutions can produce $\frac{50}{2}$ bitcoins.
- ▶ Each of the next 210,000 solutions can produce $\frac{50}{2^2}$ bitcoins.
- ▶ Each of the next 210,000 solutions can produce $\frac{50}{2^3}$ bitcoins.

Assume that this continues forever, and that bitcoins are infinitely divisible.¹ How many bitcoins can possibly be produced?

¹Actually the smallest allowed division of a bitcoin is 10^{-8} .

$$\sum_{n=0}^{\infty} 210\,000 \left(\frac{50}{2^n} \right) = 21\,000\,000$$



10 500 000



5 250 000



2 625 000



1 312 500



656 250

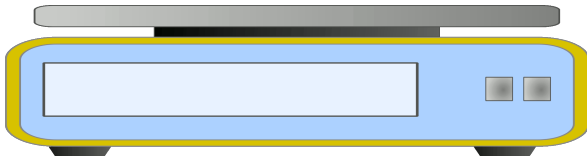
$$S_0 = 10\,500\,000$$

$$S_1 = 15\,750\,000$$

$$S_2 = 18\,375\,000$$

$$S_3 = 19\,687\,500$$

$$S_4 = 20\,343\,750$$



Arithmetic of Series

Let S , T , and C be real numbers. Let the two series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ converge to S and T respectively. Then

$$\sum_{n=1}^{\infty} [a_n + b_n] = S + T$$

$$\sum_{n=1}^{\infty} [a_n - b_n] = S - T$$

$$\sum_{n=1}^{\infty} [Ca_n] = CS$$

Geometric Series and Partial Sums

Let a and r be fixed numbers, and let N be a positive integer. Then

$$\sum_{n=0}^N ar^n = \begin{cases} a \cdot \frac{1-r^{N+1}}{1-r} & \text{if } r \neq 1 \\ a(N+1) & \text{if } r = 1 \end{cases}$$

so

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r} \text{ provided } |r| < 1$$

Evaluate $\sum_{n=0}^{\infty} \left(\frac{2}{3^n} + \frac{4}{5^n} \right)$

Geometric Series and Partial Sums

Let a and r be fixed numbers, and let N be a positive integer. Then

$$\sum_{n=0}^N ar^n = \begin{cases} a \cdot \frac{1-r^{N+1}}{1-r} & \text{if } r \neq 1 \\ a(N+1) & \text{if } r = 1 \end{cases}$$

so

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r} \text{ provided } |r| < 1$$

Evaluate $\sum_{n=6}^{\infty} \left(\frac{3^{n-1}}{5^{2n}} \right)$

Geometric Series and Partial Sums

Let a and r be fixed numbers, and let N be a positive integer. Then

$$\sum_{n=0}^N ar^n = \begin{cases} a \cdot \frac{1-r^{N+1}}{1-r} & \text{if } r \neq 1 \\ a(N+1) & \text{if } r = 1 \end{cases}$$

so

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r} \text{ provided } |r| < 1$$

Evaluate $\sum_{n=0}^{\infty} \left(\frac{2^{2n}}{3^n} \right)$

TELESCOPING SUMS

Evaluate $\sum_{n=1}^{800} \left(\frac{1}{n} - \frac{1}{n+1} \right).$

Evaluate $\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right).$

Evaluate $\sum_{n=1}^{1000} \log \left(\frac{n+1}{n} \right).$

Evaluate $\sum_{n=1}^{\infty} \log \left(\frac{n+1}{n} \right).$

Included Work

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